Measures of location and spread

This is identical to the equivalent chapter in the old S1 module. Some content will be familiar from GCSE (mean of grouped/ungrouped data), but many concepts new (e.g. standard deviation) along with possibly unfamiliar notation.

1: Mean, Median, Mode

"Calculate the mean of this grouped frequency table."

2: Quartiles, Percentiles, Deciles

"Use linear interpolation to estimate the interquartile range."

3: Variance & Standard

"Calculate the standard deviation of the maths marks."

4: Coding

"The marks x where coded using $y = 2x + 10$. Given that the standard deviation of y is 5, determine the standard deviation of x."

Variables in algebra vs stats 3

$$x$$

Similarities

- Just like in algebra, variables in stats **represent the value of some quantity**, e.g. shoe size, height, colour.
- As we saw in the previous chapter, variables can be discrete or continuous.
- **Can be part of further calculations**, e.g. if x represents height, then $2x$ represents twice people's height. In stats this is known as '**coding**', which we'll cover later.

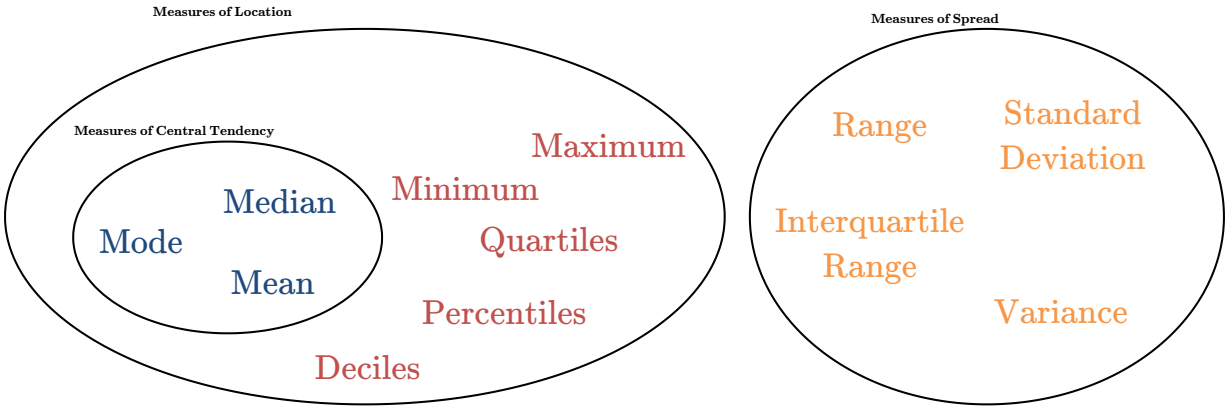
Differences

- Unlike algebra, a variable in stats represents the value of **multiple objects** (i.e. it's a bit like a set). e.g. the heights of **all** people in a room.
- Because of this, we can do **operations** on it as if it was a **collection of values**:
  - If x represents people's heights,

    $$\Sigma x$$

    gives the sum of everyone's heights. In algebra this would be meaningless: if $x = 4$, then Σx makes no sense!
  - $\overline{x}$ is the mean of x. Notice x is a collection of values whereas $\overline{x}$ is a single value.
- To each value of the variable, **we could attach an associated probability**. This is known as a **random variable** (Chapter 6).

Measure of.. 4



Measures of location are single values which describe a **position** in a data set.

Of these, **measures of central tendency** are to do with the **centre of the data**, i.e. a notion of 'average'.

Measures of spread are to do with **how data is spread out**.

x	162	178	150	160	160	170

You all know how to find the mean of a list of values. But let's consider the notation and see how theoretically we could calculate each of the individual components on a calculator.

$$\bar{x} = \frac{?}{?}$$

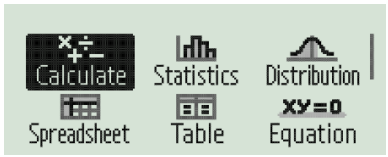
The 'overbar' in stats specifically means the sample mean of x

x	162	178	150	160	160	170

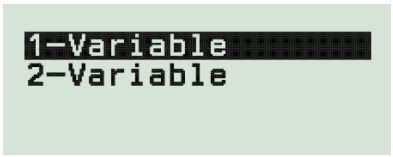
These are instructions for the **Casio fx-570/991CW**



1. Use the arrows and OK to select Statistics.



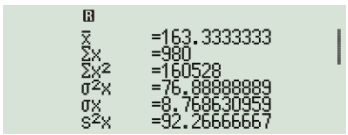
2. Choose 1-Variable.



3. Enter each value with EXE. After the last entry press OK to open the menu.



4. Select **1-Var Results** and read off the values of Σx and n and $\overline{x}$.



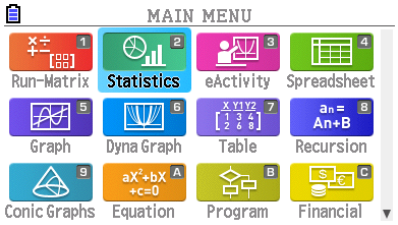
Mean and Σ x on a fx-CG50

x	162	178	150	160	160	170

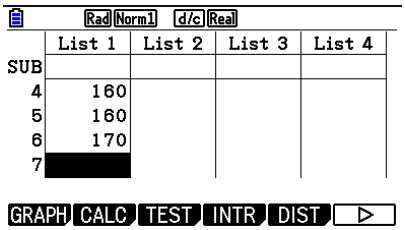
These are instructions for the **Casio fx-CG50**



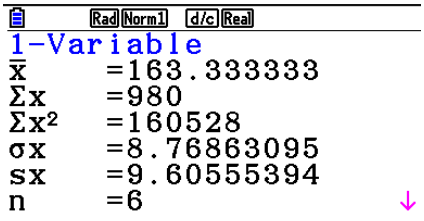
1. Choose 2 for Statistics.



2. Enter the values, pressing EXE after each.



3. Press CALC (F2) then 1-VAR (F1). Read off the values $\overline{x}$ and Σx.



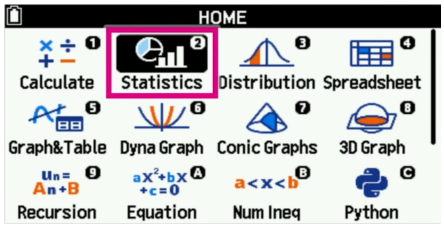
Mean and Σ x on a fx-CG100

x	162	178	150	160	160	170

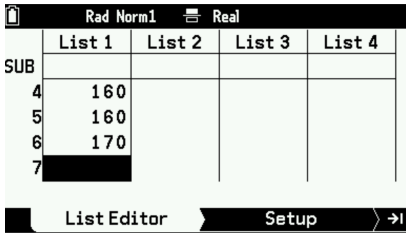
These are instructions for the **Casio fx-CG100**



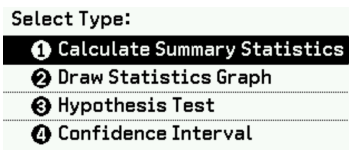
1. Choose 2 for Statistics.



2. Enter the values, pressing EXE after each.

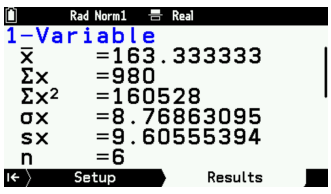


3. Select Calculate Summary Statistics, then 1-Variable. Use List 1 for XList and 1 for Freq. Select Execute.





4. Read off the values $\overline{x}$ and Σx.



Diameter of coin **x** (cm)	2.2	2.5	2.6	2.65	2.9

n ?

$\Sigma x/n$?

$3\overline{x}+1$?

Σx ?

$\overline{x}$?

7. A naturalist records the number of eggs laid by a kākāpō in each brood cycle.

Eggs	1	2	3
f	7	p	2

The mean is 1.5 eggs. Find the missing frequency.

7

The number of children x in different families is recorded.

Num children (x)	0	1	2	3	4	5
Frequency (f)	6	8	14	9	2	1

Determine the mean number of children per family, $\overline{x}$.

Since the frequency tells us how many times each value occurs, the data when listed out in full would be as follows:

0 0 0 0 0 0 1 1 1 1
1 1 1 1 2 2 2 2 2 2
2 2 2 2 2 2 2 2 3 3
3 3 3 3 3 3 3 4 4 5

The value of 2 appears 14 times. Therefore, the total of these 2's is

$$fx = 14 \times 2 = 28$$

Σfx therefore represents the total of all these products, and thus the sum of x.

The number of values is the **total frequency**, i.e. $\mathbf{n = \Sigma f}$ Therefore:

For a grouped frequency table:

$$\overline{x} = \frac{\Sigma fx}{\Sigma f}$$

The age of dogs at an animal shelter is recorded:

Age (x)	1	2	3	4	5
Frequency (f)	3	1	0	6	7

Determine the mean age $\overline{x}$.

$$\Sigma fx = (1 \times 3) + (2 \times 1) + (3 \times 0) + (4 \times 6) + (5 \times 7)$$

$$= 64$$

$$\Sigma f = 17$$

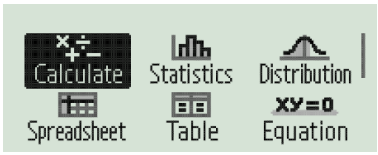
$$\overline{x} = \frac{\Sigma fx}{\Sigma f} = \frac{64}{17} = \mathbf{3.8 \text{ years}}$$

Num children (x)	0	1	2	3	4	5
Frequency (f)	6	8	14	9	2	1

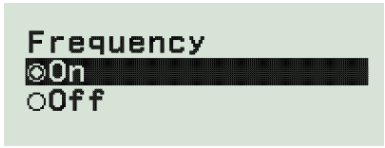
These are instructions for the **Casio fx-570/991CW**



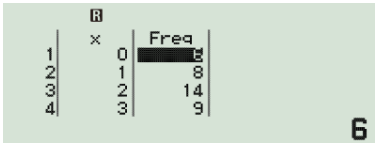
1. Use the arrows and OK to select Statistics. Choose 1-Variable.



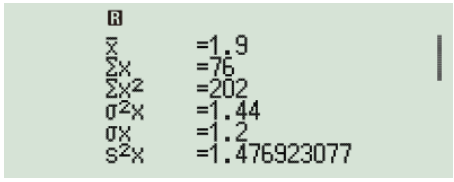
2. Press **Tools** than choose **Frequency**. Choose **On**.



3. Enter the values in x and their frequencies in Freq using EXE. Press OK for results.



4. Select **1-Var Results** and read off the values of $\overline{x}$. Note that **Σx represents Σfx** (since the x in Σx means the original data with duplicate values considered)

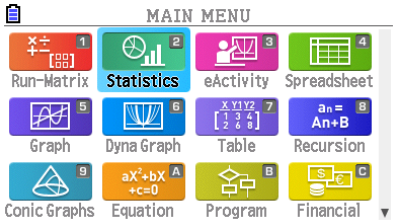


Num children (x)	0	1	2	3	4	5
Frequency (f)	6	8	14	9	2	1

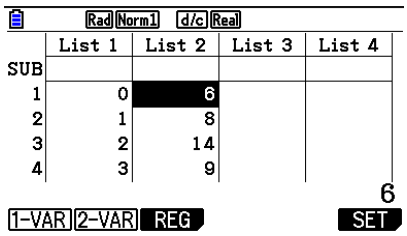
These are instructions for the **Casio fx-CG50**



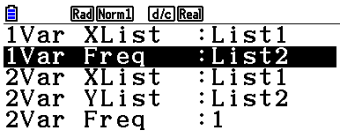
1. Choose 2 for Statistics.



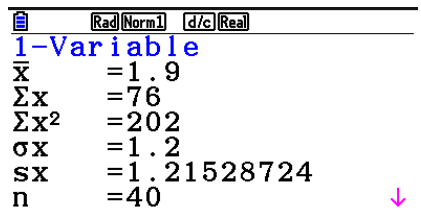
2. Enter the values of x and frequencies in the two columns, pressing EXE after each entry.



3. Press F2 (CALC), then F6 (SET). Set 1Var Freq to List 2 and press EXIT.



Press F1 for 1-Var.

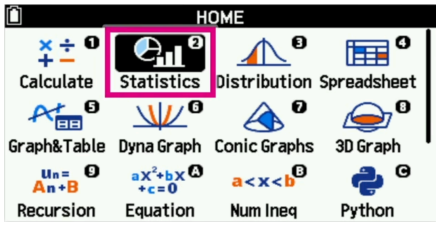


Num children (x)	0	1	2	3	4	5
Frequency (f)	6	8	14	9	2	1

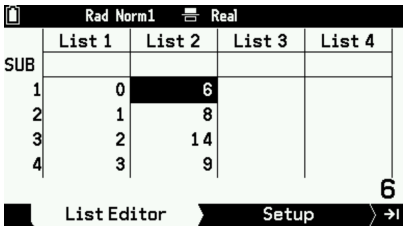
These are instructions for the **Casio fx-CG100**



1. Choose 2 for Statistics.



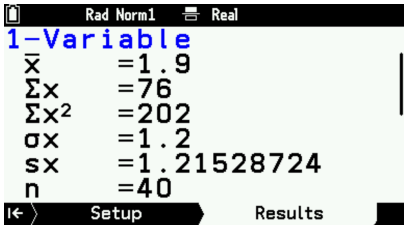
2. Enter the values of x and frequencies in the two columns, pressing EXE after each entry.



3. Select Calculate Summary Statistics, then 1-Variable. Use List 1 for XList and List 2 for Freq. Select Execute.



4. Read off the statistics



The age of members of a youth choir is recorded:

Age (x)	10	11	12	13	14	15
Frequency (f)	1	3	4	11	7	4

Determine the mean age $\overline{x}$.

?

Example: The weights of 40 randomly caught cats are summarised in the table below.

Weight (kg) x		Number of cats, f
$5-6$	5.5	8
$6-8$	7	12
$8-10$	9	15
$10-12$	11	5

Given that x denotes the midpoint of each group in the table and $\mathbf{\Sigma fx = 318}$
calculate an estimate for the mean.

If calculating from scratch:

$\Sigma fx = (8 \times 5.5) + (12 \times 7) + (15 \times 9) + (5 \times 11) = 318$

Then using the normal formula:

$$\overline{\mathbf{x}} = \frac{\mathbf{\Sigma fx}}{\mathbf{\Sigma f}} = \frac{\mathbf{318}}{\mathbf{40}} = \mathbf{7.95\ kg}$$

This time the original weights are not known, only the group each appears in.

Our 'best guess' is that each value will be, on average, the midpoint of the range it belongs to (e.g. we assume the 5-6kg cats were each 5.5kg). **We therefore replace each value with this midpoint x.**
In this way, Σfx represents the **estimated** total of the cat's weights, rather than the true (unknown) total.

Test Your Understanding 18

Actual Exam Question:
The times, in seconds, spent in a queue at a supermarket by 85 randomly selected customers, are summarised in the table below.

Time (seconds)	Number of customers, f
$0-30$	2
$30-60$	10
$60-70$	17
$70-80$	25
$80-100$	25
$100-150$	6

Given that x denotes the midpoint of each group in the table and
$\Sigma fx = 6460 \quad \Sigma fx^2 = 529400$
calculate an estimate for the mean.

(1 mark)

?

NOTE: Σfx^2 is only needed if calculating standard deviation, which is covered later.

The heights of plants, in centimetres, in a garden centre are given in the table below.

Height (cm)	Number of plants, f
$0 - 10$	7
$10 - 20$	3
$20 - 50$	8
$50 - 100$	12

Given that h denotes the midpoint of each group in the table, calculate an estimate for the mean height $\overline{h}$.

?

2. Noise levels at 30 locations near a concert were measured to the nearest decibel.

Decibels	65–69	70–74	75–79	80–84	85–89	90–94	95–99
f	1	4	6	6	8	4	1

Estimate the mean and explain why it is an estimate.

2

3. Daily mean temperatures at Heathrow in October 1987 are grouped below.

Temperature (°C)	6–8	8–10	10–12	12–14	14–16	16–18
f	3	7	9	7	3	2

Give the modal class and estimate the mean daily temperature.

3

4. Two DIY shops recorded the ages of their workers.

Age	16–25	26–35	36–45	46–55	56–65	66–75
A	5	16	14	22	26	14
B	4	12	10	28	25	13

Compare the estimated means. Which shop tends to employ older workers?

4

The table shows the heights of various cats.

Height **h** (cm)	Frequency
$0 \leq h \leq 10$	8
$10 \leq h \leq 20$	60

Estimate the number of cats with a height:

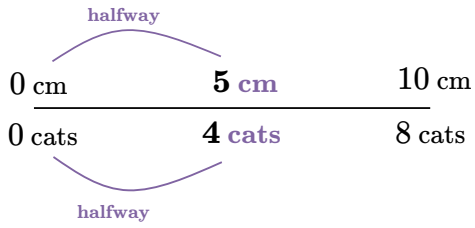
a. Below 5 cm
b. Above 15 cm
c. Below 12.5 cm

a?

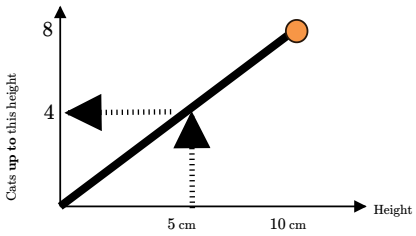
b?

c?

We are assuming the 8 cats are **equally distributed** between 0 and 10 cm, so that there are 4 between $0-5$ cm and 4 between $5-10$ cm.



This is known as **linear interpolation**, because if we were plotting this as a cumulative frequency graph (i.e. the running total of cats up to a specific height), **the graph forms a straight line**.



The grouped frequency table shows the times taken for students to complete a puzzle.

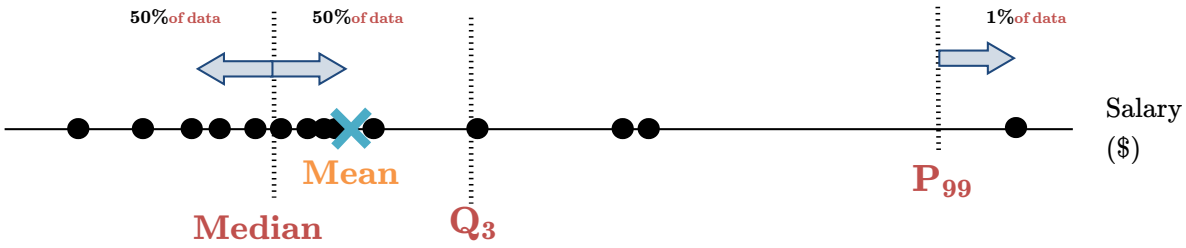
Time $\mathbf{t}$ (secs)	Frequency
$0 \leq t < 20$	12
$20 \leq h < 40$	8
$40 \leq h < 50$	20

Estimate the number of students with a time above 25 seconds.

?

Mean, Median and Measures of Position 24

MEAN: *Average* that considers all values, but is skewed by extreme values.



MEDIAN: a measure of position because it allows us to get the value a certain position, in this case 50%,along the data.

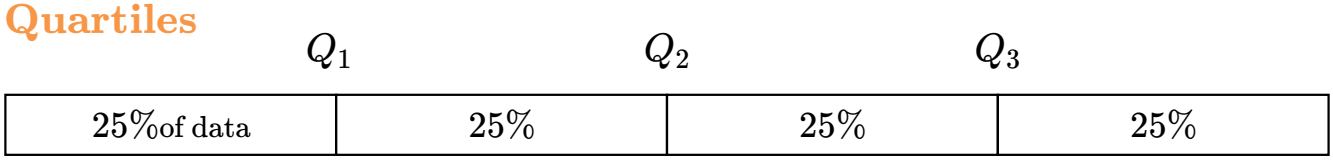
QUARTILE: gives us the value 25%, 50%, and 75%along the data.
$Q_1 = 25\%$ along data (lower quartile)
$Q_3 = 75\%$ along data (upper quartile)

PERCENTILE: % value along the data. For example, the 99th percentile (P_{99}) is the value 99%along the data. When 'the 1%'is used in the media, it's referring to people in the 'top percentile' for salary, i.e. above P_{99}.

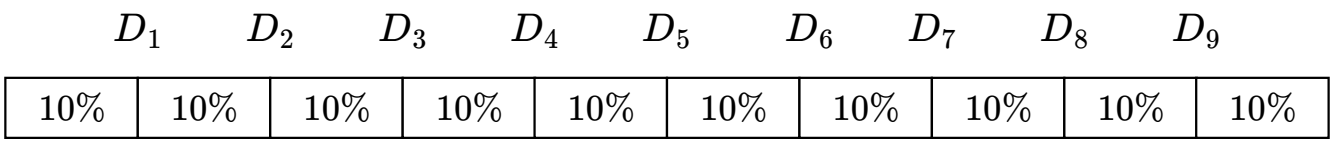
Names for Quantiles 25

Quantiles represent general positions across the data when ordered.

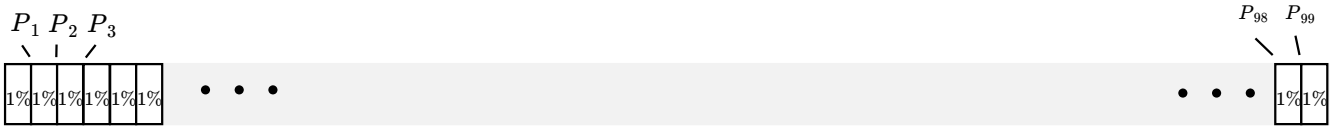
Quartiles



Deciles



Percentiles



Which Item To Use For Listed Data? 26

Items	n	Position of median	Median
$1, 4, 7, 9, 10$	5	?	?
$4, 9, 10, 15$	4	?	?
$2, 4, 5, 7, 8, 9, 11$	7	?	?
$1, 2, 3, 5, 6, 9, 9, 10, 11, 12$	10	?	?

Can you think of a rule to find the position of the median given the number of values, n?

The median is the $\frac{1}{2}(n+1)$ th item.
If the position is a decimal, use interpolation.

e.g. the '2.5^{th}' item is halfway between the 2nd and 3rd.
The '4.25^{th}' item is a quarter of the way between the 4th and 5th.

Which Item To Use For Grouped Data? 27

IQ of L6Ms2 ($\mathbf{q}$)	Frequency ($\mathbf{f}$)
$80 \leq q < 90$	7
$90 \leq q < 100$	5
$100 \leq q < 120$	3
$120 \leq q < 200$	2

If the data is grouped, what item do we use for the median?

median?

To estimate the median of grouped data, calculate $\frac{n}{2}$, then use linear interpolation.

Notice we use $\frac{n}{2}$ rather than $\frac{n+1}{2}$ if grouped data.

Quickfire Questions 28

What position do we use for the median?

1. Lengths: 3cm, 5cm, 6cm, ...
$n = 11$

Median position: **6th** 1

2. Lengths: 4m, 8m, 12.4m, ...
$n = 24$

Median position: **12th** 2

3. Age	Freq
$10 \leq a < 20$	12
$20 \leq a < 30$	5

Median position: 3

4. Score	Freq
$150 \leq s < 200$	3
$200 \leq s < 400$	7

Median position: 4

5. Ages: 5, 7, 7, 8, 9, 10, ...
$n = 60$

Median position: **30th** 5

6. Score	Freq
$150 \leq s < 200$	15
$200 \leq s < 400$	6

Median position: 6

7. Weights: 1.2kg, 3.3kg, ...
$n = 35$

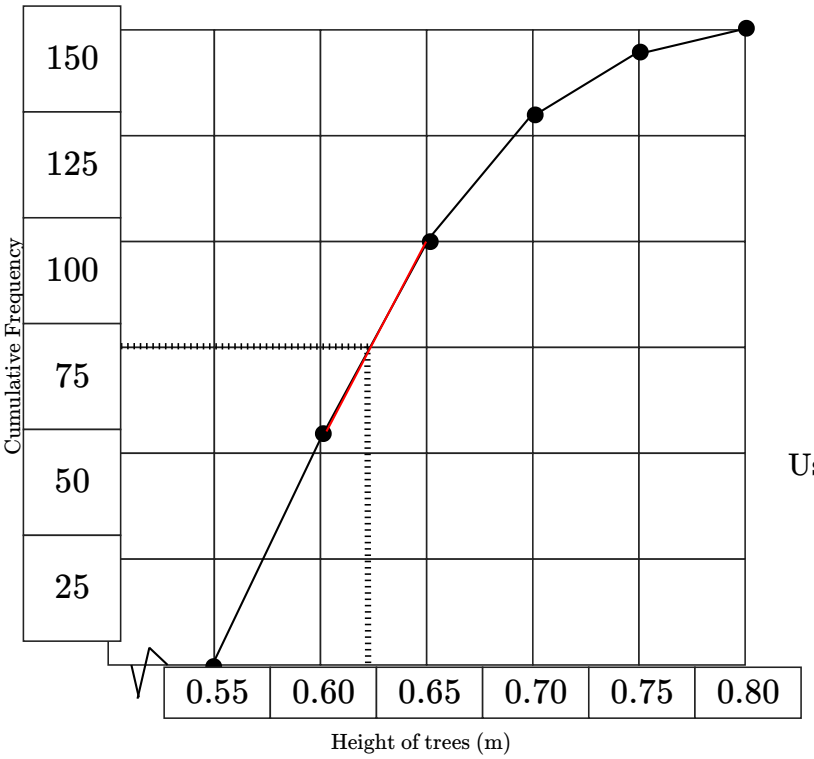
Median position: **18th** 7

8. Volume (ml)	Freq
$0 \leq v < 100$	5
$100 \leq v < 200$	6
$200 \leq v < 300$	2

Median position: 8

9. Weights: 4.4kg, 7.6kg, 7.7kg...
$n = 18$

Median position: **9th**/1 9

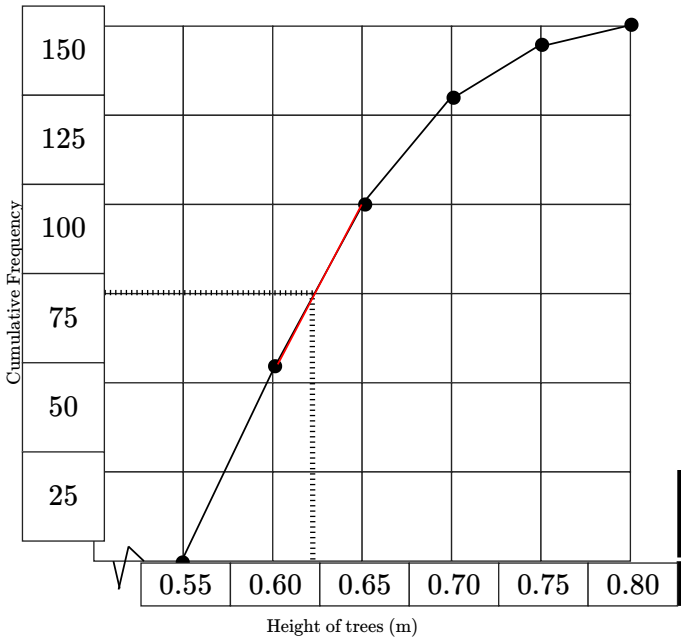


Height of tree (m)	Freq	C.F
$0.55 \leq h < 0.6$	55	55
$0.6 \leq h < 0.65$	45	100
$0.65 \leq h < 0.7$	30	130
$0.7 \leq h < 0.75$	15	145
$0.75 \leq h < 0.8$	5	150

Using a cumulative frequency graph, we know we can estimate the median by drawing a **suitable line**.

How could we read off this value exactly using a suitable calculation?

We could find the fraction of the way along the line segment using the frequencies, then go this same fraction along the class interval.



Height of tree (m)	Freq	C.F
$0.55 \leq h < 0.6$	55	55
$0.6 \leq h < 0.65$	45	100
$0.65 \leq h < 0.7$	30	130
$0.7 \leq h < 0.75$	15	145
$0.75 \leq h < 0.8$	5	150

Use linear interpolation to estimate the median.

Step 1: Identify the **interval** in which the median item, here the $\frac{150}{2} = 75^{\text{th}}$ value, lies.

Step 2: Write the relevant data for the calculation

Frequency up until this interval	Item number we're interested in?	Frequency by end of this interval?
Height at start of interval?	Quartile?	Height by end of interval?

Height of tree (m)	Freq	C.F.
$0.55 \leq h < 0.6$	55	55
$0.6 \leq h < 0.65$	45	100
$0.65 \leq h < 0.7$	30	130
$0.7 \leq h < 0.75$	15	145
$0.75 \leq h < 0.8$	5	150

What fraction of the way across the interval are we?

?

Hence:

Median = ?

Frequency up until this interval → 55

Item number we're interested in. → 75

Frequency by end of this interval → 100

0.6m Med 0.65m

Height at start of interval.

Height by end of interval.

Tip: put the units to avoid getting frequencies confused with values of the variable.

Tip: To quickly get frequency before and after, just look for the two cumulative frequencies that surround the item number.

1. Weight of cat (kg)	Freq	C.F.
$1.5 \leq w < 3$	10	10
$3 \leq w < 4$	8	18
$4 \leq w < 6$	14	32

Median class interval: ?

? ? ?

? Med ?

Fraction along interval: ?

Median: ?

2. Time (s)	Freq	C.F.
$8 \leq t < 10$	4	4
$10 \leq t < 12$	3	7
$12 \leq t < 14$	13	20

? ? ?

? Med ?

Median:

?

What's different about the interval here? 33

Weight of cat to nearest kg	Frequency
$10 - 12$	7
$13 - 15$	2
$16 - 18$	9
$19 - 20$	4

There are **GAPS** between intervals!
What interval does this **actually** represent?

$$10 - 12$$

Lower class boundary? — upper class boundary?

Class Width?

Identify the class width 34

1. Distance **d** travelled (in m)	...
$0 \leq d < 150$	
$150 \leq d < 200$	
$\mathbf{200 \leq d < 210}$	

Lower class boundary = ?

Class width = ?

2. Weight **w** in kg	...
$10 - 20$	
$21 - 30$	
$\mathbf{31 - 40}$	

Lower class boundary = ?

Class width = ?

3. Time **t** taken (in seconds)	...
$0 - 3$	
$\mathbf{4 - 6}$	
$7 - 11$	

Lower class boundary = ?

Class width = ?

4. Speed **s** (in mph)	...
$10 \leq s < 20$	
$20 \leq s < 29$	
$\mathbf{29 \leq s < 31}$	

Lower class boundary = ?

Class width = ?

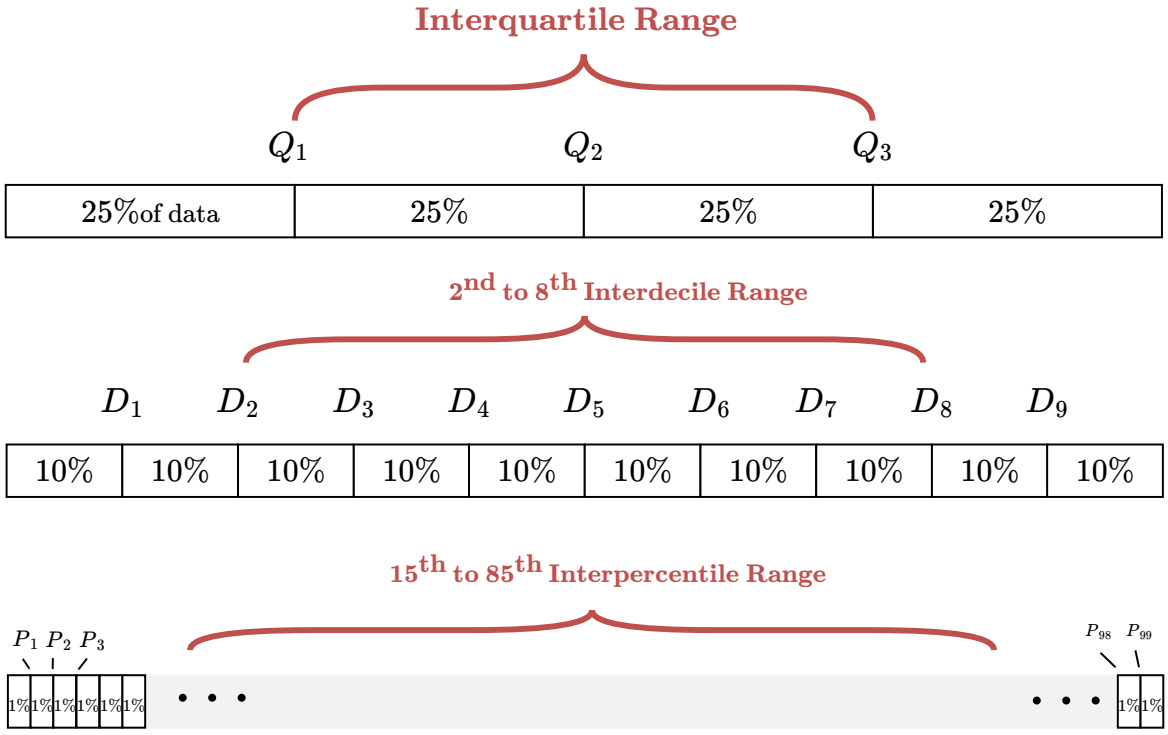
Example:

Summarised below are the distances, to the nearest mile, travelled to work by a random sample of 120 commuters.

Use linear interpolation to estimate the distribution's median

Distance (to the nearest mile)	Number of commuters	
0 – 9	10	10
10 – 19	19	29
20 – 29	43	72
30 – 39	25	97
40 – 49	8	105
50 – 59	6	111
60 – 69	5	116
70 – 79	3	119
80 – 89	1	120

$$Median = 1^{?} \quad \frac{^{?} \quad ^{?} \quad ^{?}}{^{?} \quad \text{Med} \quad ^{?}}$$



Just as we can calculate the interquartile range to mean the range of the middle 50% of data, we can also use deciles and percentiles to find the range of a more general middle percentage of the data.

Use linear interpolation to estimate the 3rd to 7th interdecile range of the following data:

Mass (x) to the nearest gram	Freq ($\mathbf{f}$)
$10 - 12$	5
$13 - 14$	15
$15 - 18$	7
$19 - 22$	3

?

3. A hotel records the number of lift breakdowns each week for 26 weeks.

Breakdowns	0–1	2–3	4–5
f	18	7	1

Use interpolation to estimate the median number of breakdowns. (2 marks)

3

1. Estimate the median daily attendance at a hospital trauma clinic.

Patients	10–19	20–29	30–34	35–39	40–44	45–49	50–69
f	2	18	24	30	27	14	5

1

2. Use interpolation to estimate the median age of the 300 houses.

Age (years)	0–20	20–40	40–60	60–100	100–200	200–300	300–500
f	36	92	74	39	14	27	18

2

3. Estimate the median length of a visit to the cyber-café.

Stay (min)	0–30	30–60	60–90	90–120	120–240	240–360
f	15	31	32	23	17	2

3

4. The following table summarises the times, t minutes to the nearest minute, recorded for a group of students to complete an exam.

Minutes	11–20	21–25	26–30	31–35	36–45	46–60
f	62	88	16	13	11	10

a) Estimate the mean of this data

b) Use linear Interpolation to estimate the value of the median

a?

b?

You need to be able to find the **quartiles** of both listed data and of grouped data. The rule is exactly the same as for the median.

Listed data

Items	**n**	Position of LQ & UQ	LQ & UQ
$1, 4, 7, 9, 10$	5	?	?
$4, 9, 10, 15$	4	?	?
$2, 4, 5, 7, 8, 9, 11$	7	?	?
$1, 2, 3, 5, 6, 9, 9, 10, 11, 12$	10	?	?

Can you think of a rule to find the position of the LQ/UQ given n?

?

Grouped data

IQ of L6Ms2 ($\mathbf{q}$)	Frequency ($\mathbf{f}$)
$80 \leq q < 90$	7
$90 \leq q < 100$	5
$100 \leq q < 120$	3
$120 \leq q < 200$	2

Position to use for LQ:

?

To find the LQ and UQ of grouped data, find $\frac{1}{4}n$ and $\frac{3}{4}n$, then use linear interpolation.

Again, **DO NOT** round this value.

The LQ, median and UQ give you 25%, 50% and 75% along the data respectively.

$n = 43$

Item to use for 57th percentile?



?

You will always find these for grouped data in an exam, **so never round this position**.

Notation:

Lower Quartile: ?

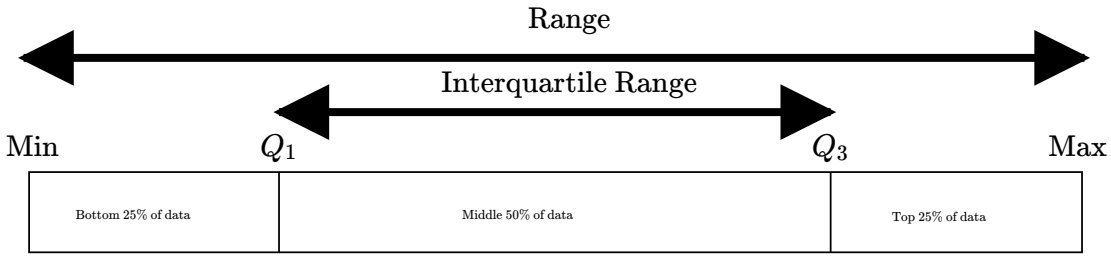
Median: ?

Upper Quartile: ?

57th Percentile: ?

Measure of spread

The interquartile range and interpercentile range are examples of **measures of spread.**



Interquartile Range = Upper Quartile − Lower Quartile

Why might we favour the interquartile range over the range?

?

Test your understanding

These are the same as the 'Test Your Understanding' questions on your supplementary sheet from before.

Age of relic (years)	Frequency
0-1000	24
1001-1500	29
1501-1700	12
1701-2000	35

Item for Q_1:

?

?

?

?

Shark length (cm)	Frequency
$40 \leq x < 100$	17
$100 \leq x < 300$	5
$300 \leq x < 600$	8
$600 \leq x < 1000$	11

Item to use for P_{10}:

?

?

?

10th to 90th interpercentile range:

?

5. A roadside assistance firm recorded customer waiting times.

Minutes	20–30	30–40	40–50	50–60	60–70
f	6	10	18	13	2

(a) Estimate the mean waiting time. (1 mark)

(b) Calculate the 65th percentile. (2 marks)

(c) Only 10% of customers wait longer than 56 minutes, the firm claims. Calculate a suitable percentile and comment. (3 marks)

5

6. The wingspans of 100 Californian condors are recorded in metres.

Wingspan	1–1.5	1.5–2	2–2.5	2.5–3	3+
f	4	20	37	28	11

Estimate and interpret the 80th percentile. Why can the 90th percentile not be estimated by interpolation?

6

2. Monthly incomes of factory workers are grouped in pounds.

Income (£)	900–1000	1000–1100	1100–1200	1200–1300
f	3	24	28	15

Estimate the 34th to 66th interpercentile range and the number of incomes in it.

2

3. Lancaster to Preston journey times were recorded to the nearest minute.

Minutes	15–16	17–18	19–20	21–22
f	5	10	35	10

Estimate the 5th to 95th interpercentile range and the number of journeys within it.

3

4. Daily mean temperatures in Leeming for the first ten days of June 1987 (°C):
14.3, 12.7, 12.4, 10.9, 9.4, 13.2, 12.1, 10.3, 10.3, 10.6
Find the median and IQR. Compare with May (median 9.9°C, IQR 3.9°C). In July the 10th to 90th interpercentile range was 5.4°C. Estimate how many days fell in that range.

4

1. Exam completion times were recorded to the nearest minute.

Minutes	11–20	21–25	26–30	31–35	36–45	46–60
f	62	88	16	13	11	10

(c) Show that the lower quartile is 18.6 minutes. (d) Estimate the IQR.

1

2. Distances travelled to London by 134 examiners, to the nearest km:

Distance (km)	41–45	46–50	51–60	61–70	71–90	91–150
f	4	19	53	37	15	6

Use interpolation to estimate the median and both quartiles.

2

3. Estimate both quartiles and the IQR for the ages of 300 houses.

Age (years)	0–20	20–40	40–60	60–100	100–200	200–300	300–500
f	36	92	74	39	14	27	18

3

4. Estimate both quartiles and the 90th percentile for café visits.

Stay (min)	0–30	30–60	60–90	90–120	120–240	240–360
f	15	31	32	23	17	2

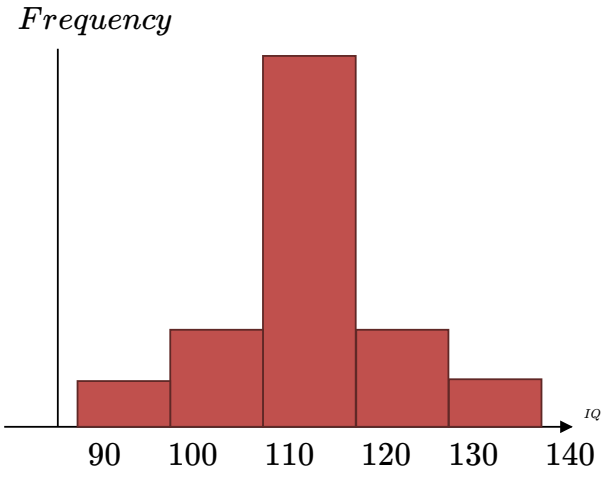
4

5. Commuter distances, to the nearest mile. Estimate the IQR.

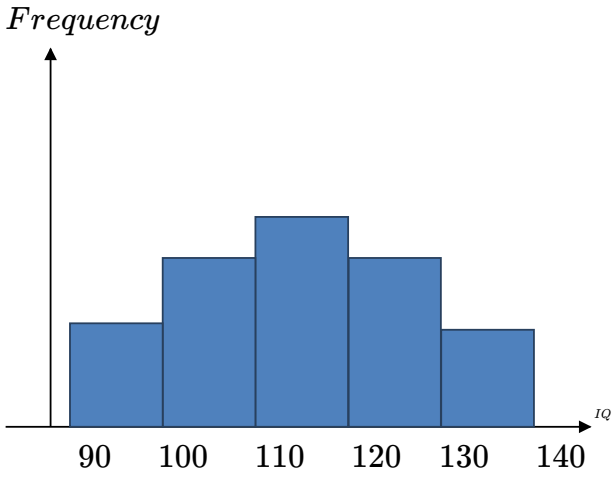
Miles	0–9	10–19	20–29	30–39	40–49	50–59	60–69	70–79	80–89
f	10	19	43	25	8	6	5	3	1

5

Distribution of IQs in Class A



Distribution of IQs in Class B



Here are the distribution of IQs in two classes of the same size. What's the same, and what's different?

?

Variance 53

Variance is a measure of spread that takes all values into account.
Variance, by definition, is the **average squared distance from the mean.**

$$\sigma^2 = \frac{\sum(x-\bar{x})^2}{n}$$

Notation: While Σ is 'uppercase sigma' and means 'sum of', σ is 'lowercase sigma' (we'll see why we have the squared in a sec)

Distance from mean. . .

Squared distance from mean. . .

Average squared distance from mean. . .

Simpler formula for variance 54

In practice, it is possible to simplify the formula to the following*:

Variance mean of the squares minus the square of the mean:

$$\sigma^2 = \boxed{?} - \boxed{?}$$

Standard Deviation

$$\sigma = \sqrt{Variance}$$

*** Proof: (not in syllabus)**

?

The standard deviation **can 'roughly' be thought of as the average distance from the mean.**

3, 11

Variance

?

Standard deviation

?

2cm 3cm 3cm 5cm 7cm

Variance

?

Standard deviation

?

Find the variance and standard deviation of the following sets of data.

2 4 6

Variance = ?

Standard Deviation = ?

1 2 3 4 5

Variance = ?

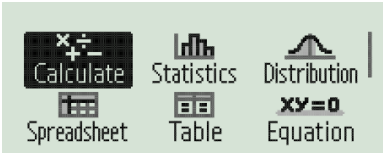
Standard Deviation = ?

Time (secs)	30	34	35	39

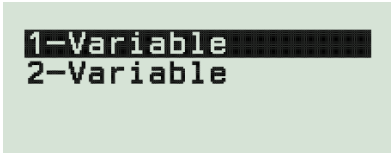
These are instructions for the **Casio fx-570/991CW**



1. Use the arrows and OK to select Statistics.



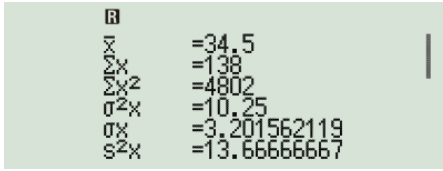
2. Choose 1-Variable.



3. Enter each value with EXE. After the last entry press OK to open the menu.



4. Choose '1 Variable Results'. Read off σx (σ) and $\sigma^2 x$ (the variance σ^2)

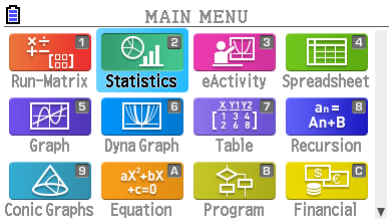


Time (secs)	30	34	35	39

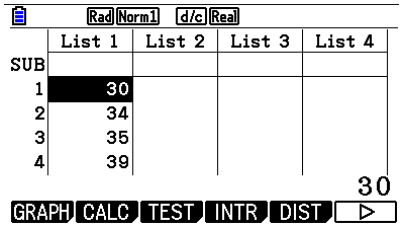
These are instructions for the **Casio fx-CG50**



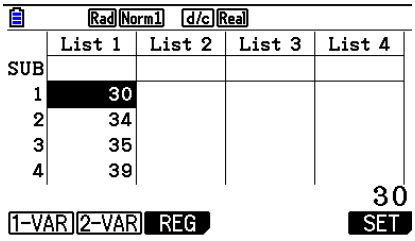
1. Choose 2 for Statistics.



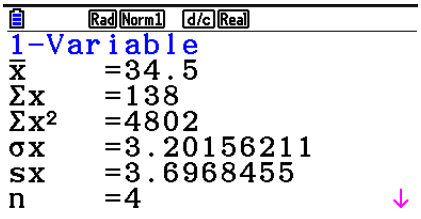
2. Enter your values, pressing EXE after each and using the arrow keys to move around.



3. In CALC (F2), open SET (F6). Set 1Var Freq to 1, press EXIT, then 1-VAR (F1).



4. Read off σ.

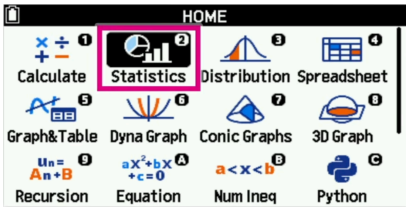


Time (secs)	30	34	35	39

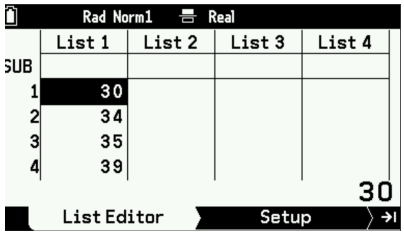
These are instructions for the **Casio fx-CG100**



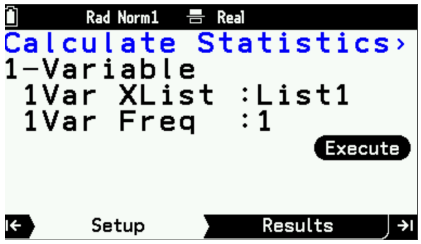
1. Choose 2 for Statistics.



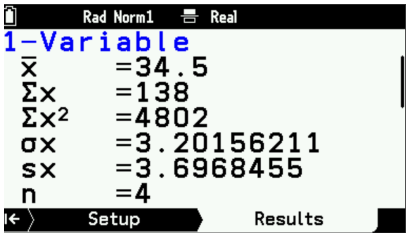
2. Enter your values, pressing EXE after each and using the arrow keys to move around.



3. Select Calculate Summary Statistics, then 1-Variable. Use List 1 for XList and 1 for Freq. Select Execute.



4. Read off σ.



Use a calculator to determine the standard deviation of the following data set of long jump distances.

6.1 m, 8.3 m, 8.3 m, 8.4 m, 8.5 m, 8.7 m

?

Calculating σ Using Given Summations 61

It is unlikely in an exam you would be asked to compute the standard deviation from scratch. Most likely you would calculate σ directly without formula or be given key summations to use in the formula.

EXAMPLE

Before going on holiday to Seapron, Tania records the weekly rainfall (x mm) at Seapron for 8 weeks during the summer. Her results are summarised as

$$\Sigma x = 86.8 \qquad \Sigma x^2 = 985.88$$

$$\sigma^2 = \frac{\Sigma x^2}{n} - \overline{x}^2$$

Find the standard deviation, σ_x, for these data.

(3 marks)

1. Calculate $\overline{x}$ using $\frac{\Sigma x}{n}$:

\overline{x}?

2. Use the formula with provided Σx^2:

{σ}^{2}?

3. Calculate σ_x :

σ?

Test Your Understanding 62

[Edexcel AS SAM P2 Q1c]

Sara is investigating the variation in daily maximum gust, t kn, for Camborne in June and July 1987. She used the large data set to select a sample of size 20 from the June and July data for 1987.

Sara selected the first value using a random number from 1 to 4 and then selected every third value after that.

The data Sara collected are summarised as follows

$$\mathrm{n} = 20 \qquad \Sigma \mathrm{t} = 374 \qquad \Sigma \mathrm{t}^2 = 7600$$

Calculate the standard deviation.

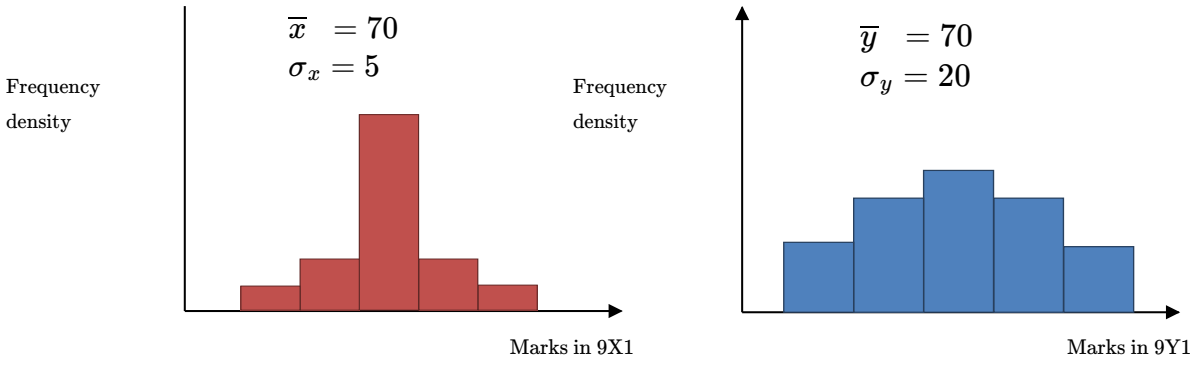
(2 marks)

Mark scheme

You may want to use the following mnemonic:

Variance: "The mean of the squares minus the square of the mean ('msmsm')"

$$\text{Variance} = \boxed{?} - \boxed{?}$$



Compare the marks of the two classes.

The classes had the same mean mark, but 9X1 had a lower standard deviation, suggesting the marks were **more consistent**.

Note: When σ is lower, it means there's less variation, which means **more consistency**.

Previously when comparing two data sets, we might compare:

1. A measure of **central tendency** (usually the mean)
2. A measure of **spread**

For the latter, you may have previously used the range or interquartile range. **Now you can use the standard deviation**!

Exam Question

The number of hours of sunshine each day, y, for the month of July at Heathrow are summarised in the table below.

Hours y	$0 \leq y < 5$	$5 \leq y < 8$	$8 \leq y < 11$	$11 \leq y < 12$	$12 \leq y < 14$
f	12	6	8	3	2

It can be shown that the mean of this data is 6.63 and that the standard deviation is 3.69.

The mean and standard deviation for the number of hours of daily sunshine for the same month in Hurn are 5.98 hours and 4.12 hours respectively.

Thomas believes that the further south you are the more consistent should be the number of hours of daily sunshine.

State whether or not the standard deviations for the two locations support Thomas' belief.

Mark Scheme?

Effect of an Added Value on σ 66

If we consider the standard deviation as, approximately, the average distance of each value from the mean, we can reflect on how additional values will impact the standard deviation.

The populations of 10 countries are recorded. The mean population is 9 million, and the standard deviation is 1 million.

I add Switzerland to my data set, which has a population of 9 million.

State the effect on:

a. the range of the data

b. the standard deviation of the data

a?

b?

Standard Deviation From Grouped Data 67

Data is...	Mean	Variance
Ungrouped	$\overline{x} = \frac{\Sigma x}{n}$	$\sigma^2 = \frac{\Sigma x^2}{n} - \overline{x}^2$
Grouped	?	?

MEAN
If the data is grouped, then we previously saw that to calculate the mean, fx gives the total when x is written out f times, so accounts for the duplication of values.
$\Sigma f = n$, because the total frequency is the number of data values.

VARIANCE
Each value is duplicated f times, so each squared value will also be duplicated f times, contributing fx^2 to the total.

Example With Given Summations 68

EXAM QUESTION
The following table summarises the times, t, minutes to the nearest minute, recorded for a group of students to complete an exam.

Time t (min)	11–20	21–25	26–30	31–35	36–45	46–60
Students f	62	88	16	13	11	10

[You may use $\Sigma ft^2 = 134281.25$*]*

Estimate the mean and standard deviation of these data. **(5)**

1. Find the class midpoints.

2. Estimate the mean.

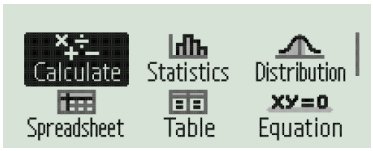
3. Estimate the standard deviation.

Time (minutes) t	$11-20$	$21-25$	$26-30$	$31-35$	$36-45$	$46-60$
Midpoints	15.5	23	28	33	40.5	53
Number of students f	62	88	16	13	11	10

These are instructions for the **Casio fx-570/991CW**



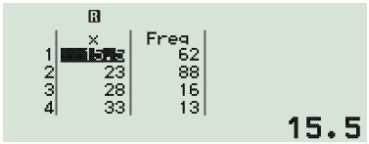
1. Use the arrows and OK to select Statistics.



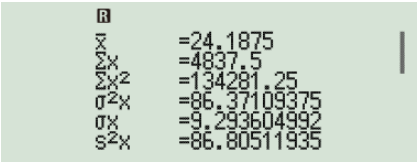
2. Choose 1-Variable. Open Tools, set Frequency to On, then press Back.



3. Enter the values in x and their frequencies in Freq using EXE. Press OK for results.



4. Choose '1 Variable Results'. Read off σx (σ) and $\sigma^2 x$ (the variance σ^2)



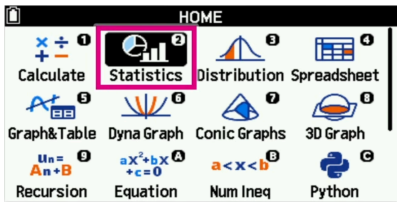
Important note: Σx^2 here means $\Sigma f x^2$, consistent with the value given in the question on the previous slide. The frequencies are implicitly 'baked in'.

Time (minutes) t	$11-20$	$21-25$	$26-30$	$31-35$	$36-45$	$46-60$
Midpoints	15.5	23	28	33	40.5	53
Number of students f	62	88	16	13	11	10

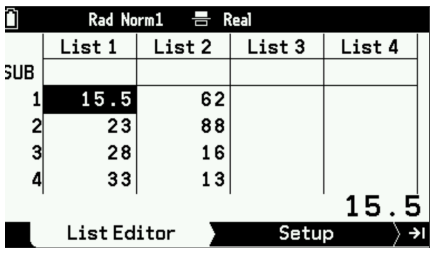
These are instructions for the **Casio fx-CG100**



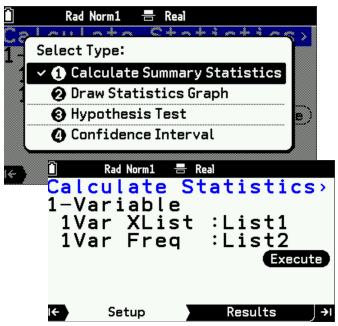
1. Choose 2 for Statistics.



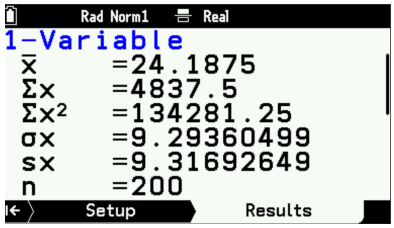
2. Enter the midpoints in List 1 and the Frequencies in List 2



3. Select Calculate Summary Statistics, then 1-Variable. Use List 1 for XList and List 2 for Freq. Select Execute.



4.Read off σx (σ). Square this value for the variance.



Exam Question

An agriculturalist is studying the yields, y kg, from tomato plants. The data from a random sample of 70 tomato plants is summarised below.

Yield (y kg)	**Frequency** (f)	**Yield midpoint** (x kg)
$0 \leq y < 5$	16	2.5
$5 \leq y < 10$	24	7.5
$10 \leq y < 15$	14	12.5
$15 \leq y < 25$	12	20
$25 \leq y < 35$	4	30

(You may use $\Sigma fx = 755$ and $\Sigma fx^2 = 12037.5$)

Estimate the mean and the standard deviation of the yields of the tomato plants.

Mark Scheme?

Example: In classes 7A and 7B, consisting of 20 and 30 students respectively, the sums of their maths scores were $\Sigma a = 1200$ and $\Sigma b = 2100$, and $\Sigma a^2 = 75000$ and $\Sigma b^2 = 160000$.
Determine the standard deviation of maths scores in the two classes combined.

Let x be the scores across both classes.

The number of values, n, is just the total students across the two sets.

n?

Similarly, the total score, Σx, across both classes is just the sum from each class:

\mathrm{\Sigma }x and \mathrm{\Sigma }{x}^{2}?

Once we have n, Σx and Σx^2 across the combined set, calculate the standard deviation:

σ?

Further Example

In classes 8A and 8B, consisting of 5 and 10 students respectively, the means of their maths scores were 60 and 70, and the standard deviations 6 and 7.
Determine the standard deviation across all 15 students.

1. Use $\overline{x} = \frac{\Sigma x}{n}$ formula to determine Σx

Σa?

2. Similarly use σ^2 formula backwards to determine Σx^2

Σa²?

3. Repeat the above steps for class 8B

Σb and Σb²?

4. combine sums together.

σ?

Test Your Understanding

Exam Question

Throughout the year Alice took nine written tests and five practical tests in Music.

The mean mark of her nine written tests was 69.

The mean mark of her five practical tests was 83.

The standard deviation of her nine written tests was 15.4.

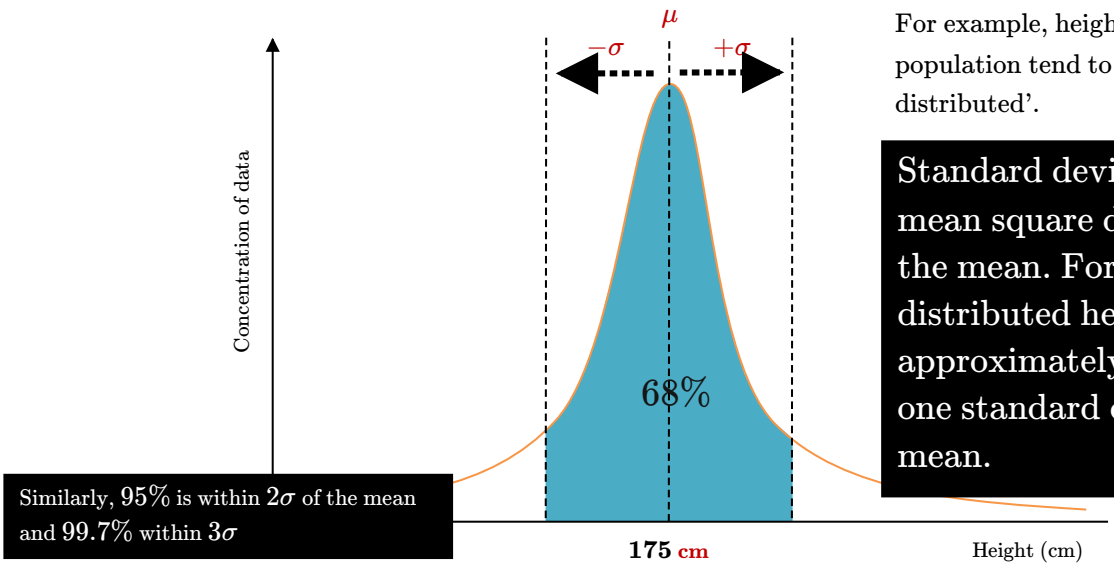
The standard deviation of her five practical tests was 7.8.

Calculate the standard deviation of all 14 tests.

(4 marks)

?

Standard deviation can be used as a measure of spread **for data with any 'shape'**. However, it becomes **particularly important when we consider normally distributed data**, i.e. has a symmetrical 'bell-curved' shape with data becoming less concentrated further away from the mean.



For example, heights in a population tend to be 'normally distributed'.

Standard deviation is the root mean square distance from the mean. For normally distributed heights, approximately 68% lie within one standard deviation of the mean.

Similarly, 95% is within 2σ of the mean and 99.7% within 3σ

You can find out about the normal distribution in skills 640-645.

2. Ten collie dogs are weighed. Use the summaries to find the standard deviation.

$\sum w = 241, \qquad \sum w^2 = 5905$

2

4. Find the mean and standard deviation for the two sets combined.

Set	n	$\sum x$	$\sum x^2$
1	10	50	310
2	15	86	568

4

5. Weekly pocket money was recorded to the nearest pound.

Pounds	8	9	10	11	12
f	14	8	28	15	20

Find the mean and standard deviation. Estimate how many received more than one standard deviation above the mean.

5

6. Calculate the standard deviation of days absent in one term.

Days absent	0	1	2	3	4
f	12	20	10	7	5

6

7. Lifetimes of 50 machine parts are grouped by hours (upper endpoint included).

Hours	5–10	10–15	15–20	20–25	25–30
f	5	14	23	6	2

Assess the claim that 90% lasted longer than one standard deviation below the mean.

7

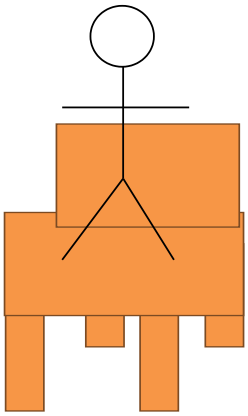
8. Daily mean windspeeds in Leeming, June 2015, ranged from 4 to 17 knots.

$n = 30, \quad \sum x = 243, \quad \sum x^2 = 2317$

Find the mean and standard deviation. Estimate how many days exceeded the mean by more than one standard deviation and state your assumption.

8

What do you reckon is the mean height of people in a room if they stand on a chair?



Is there an easy way to recalculate the mean based on the new heights? And the variance of the heights?

?

Suppose now after a bout of 'stretching you to your limits', you're now all 3 times your original height.

What do you think happens to the standard deviation of your heights?

?

What do you think happens to the variance of your heights?

?

Extension Question: Can you prove the latter using the formula for variance?

?

Suppose our original variable (e.g. heights in cm) was x.
Then y would represent the heights with 10cm added on to each value.

Coding	Effect on $\overline{x}$	Effect on σ
$y = x + 10$	?	?
$y = 3x$	?	?
$y = 2x - 5$	?	?

The point of coding

Cost x of diamond ring (£)

£1010 £1020 £1030 £1040 £1050

We 'code' our variable using the following:

$$y = \frac{x-1000}{10}$$

New values y:

?

Standard deviation of y (σ_y): ?

therefore...

Standard deviation of x (σ_x): ?

Quickfire questions

Old mean $\overline{x}$	Old σ_x	Coding	New mean $\overline{y}$	New σ_y
36	4	$y = x - 20$	?	?
?	?	$y = 2x$	72	16
35	4	$y = 3x - 20$	?	?
?	?	$y = \frac{x}{2}$	20	$\frac{3}{2}$
11	27	$y = \frac{x+10}{3}$	?	?
?	?	$y = \frac{x-100}{5}$	40	5

The following table summarises the times, t minutes to the nearest minute, recorded for a group of students to complete an exam.

Time (minutes) t	11 – 20	21 – 25	26 – 30	31 – 35	36 – 45	46 – 60
Number of students f	62	88	16	13	11	10

[You may use $\sum ft^2 = 134281.25$]

(*a*) Estimate the mean and standard deviation of these data. **(5)**
(*b*) Use linear interpolation to estimate the value of the median. **(2)**
(*c*) Show that the estimated value of the lower quartile is 18.6 to 3 significant figures. **(1)**
(*d*) Estimate the interquartile range of this distribution. **(2)**
(*e*) Give a reason why the mean and standard deviation are not the most appropriate summary statistics to use with these data. **(1)**

Suppose we've worked all these out already.

The person timing the exam made an error and each student actually took 5 minutes less than the times recorded above. The table below summarises the actual times.

Time (minutes) t	6 – 15	16 – 20	21 – 25	26 – 30	31 – 40	41 – 55
Number of students f	62	88	16	13	11	10

(*f*) Without further calculations, explain the effect this would have on each of the estimates found in parts (*a*), (*b*), (*c*) and (*d*). **(3)**

f)

?

3. Television prices are coded as follows. Find the original mean price.

$y = (x - 65)/200, \qquad \bar{y} = 1.5$

3

6. Weekly incomes of 100 workers are coded. Find the standard deviation of their original incomes.

$y = (i - 90)/100, \quad \sum y = 131, \quad \sum y^2 = 176.84$

6

7. Annual rainfall at six places is coded. Estimate the standard deviation in millimetres.

$s = 0.01x - 10, \quad \sum s = 16.1, \quad \sum s^2 = 147.03$

7

9. Pressures in Hurn in June 2015 are coded. Find the original mean and standard deviation.

$t = p/2 - 500, \quad n = 30, \quad \bar{t} = 10.15, \quad S_{tt} = 296.4$

9

8. A teacher adds 12 to each mark and then reduces the result by 20%. Write a formula relating the standardised mark to the original mark. Find the original mean and standard deviation given:

$n = 28, \qquad \bar{t} = 52.8, \qquad S_{tt} = 7.3$

8

I have a list of 30 heights in the class. What item do I use for:

Q1 ?

Q2 ?

Q3 ?

For the following grouped frequency table, calculate:

Height $\mathbf{h}$ of bear (in metres)	Frequency
$0 \leq h < 0.5$	4
$0.5 \leq h < 1.2$	20
$1.2 \leq h < 1.5$	5
$1.5 \leq h < 2.5$	11

a) The estimate mean: ?

b) The estimate median: ?

c) The estimate variance:
(you're given $\Sigma f h^2 = 67.8125$) ?

What is the standard deviation of the following lengths: 1cm, 2cm, 3cm

?

The mean of a variable x is 11 and the variance 4.
The variable is coded using $y = \frac{x+10}{3}$. What is:

a) The mean of y? ?

b) The variance of y? ?

A variable x is coded using $y = 4x - 5$.
For this new variable y, the mean is 15 and the standard deviation 8.
What is:

c) The mean of the original data? ?

d) The standard deviation of the original data? ?